Transparency in Music-Generative AI: A Systematic Literature Review

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Research Article

Keywords: AI Transparency, Music Generation, Systematic Literature Review, Generative AI, Music, Ethics AI

Posted Date: December 17th, 2024

DOI: https://doi.org/10.21203/rs.3.rs-3708077/v2

Additional Declarations: The authors declare no competing interests.

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Recent advancements in music-generative AI raise ethical, social, legal and economic concerns linked to artists' work, the existing music industry model, the role of AI in creative processes, and intellectual property rights. Transparency, a pillar for trustworthy AI, is key to addressing the principal ethical implications of generative AI in the music domain. We analyse transparency approaches for generative AI in music with a two-stage systematic literature review, identifying 107 relevant publications. Findings reveal a growing interest in AI transparency and the ethical implications of generative models. Yet, transparent methodologies for music-generative AI remain an under-explored topic, highlighting such research gap and the need to expand research in this direction. To encourage future exploration, we created a dynamic list of relevant publications in a public repository to be updated with new research initiatives.

CCS Concepts: • **Applied computing → Sound and music computing**; • **General and reference → Surveys and overviews**.

Additional Key Words and Phrases: Transparency, Music Generation, AI-Generated Music, Ethics, Systematic Literature Review

ACM Reference Format:
Roser Batlle-Roca, Emilia Gómez, Wei-Hsiang Liao, Xavier Serra, and Yuki Mitsufuji. 2024. Transparency in Music-Generative AI: A Systematic Literature Review. 1, 1 (December 2024), 33 pages. https://doi.org/XXXXXXX.XXXXXXX

1 INTRODUCTION

Despite a long existing research tradition on computational methods in algorithmic composition and automatic music generation [58, 69, 103], recent advancements in artificial intelligence (AI) are transforming and increasing the popularity of music generation algorithms. These advancements are encouraging their deployment in real-world scenarios. Notably, several influential music generation models with strong support from the industry were released between 2023 and 2024, such as *MusicLM* (Google Research), *MusicGen* (Meta AI), *Stable Audio* (Stability AI), *Suno* (Suno AI) and *Udio* (Udio), further underscoring the rapid advancements in this field. However, multiple concerns are arising about the potential ethical, social, legal and economic implications of these new AI-powered systems, which are linked to how AI might replace or complement human creative tasks [36, 48], the uncertain and opaque nature of generative algorithms [32, 128], and the many challenges involved in the objective evaluation of both human and AI-generated works [6, 132].

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Manuscript submitted to ACM

Music generation strategies are commonly categorised into the symbolic and audio domains [131]. Many models rely on MIDI files or audio databases, but recent advancements have introduced novel models that explore generating music from text prompts (e.g., *MusicGen* [44] and *Stable Audio* [55]) or even images (e.g., *BGT-G2G* [130]). In these cases, text or images are applied as input to guide or control the generation process, expanding the creative possibilities of AI in the music domain. Despite the advancements in music generation, many research initiatives often overlook the potential ethical negative consequences of their systems and generated outputs [16]. Most concerns focus on the use of computational tools to generate creative music behaviours [32], the effects on creative tasks traditionally undertaken by musicians, performers, and producers, along with implications for business models and listening experiences, and the connection to copyright and intellectual property (IP) management [66, 123]. Nowadays, there is an increasing discussion regarding the potential replication of music samples and plagiarism of the training set, the originality of AI-generated music, and the lack of evaluation strategies of both issues [123, 134]. This involves questioning whether the generative model retrieves and copies fragments from the input songs and whether the generated music can be considered original. Determining whether AI-generated music is original (or not) is a critical detail considering copyright regulation [43]. This specific challenge is motivating a new research trend, leading to novel methodologies for assessing attribution, deepfakes and replication in AI-generated music, such as the ones introduced by Afchar et al. (2024) [4], Barnett et al. (2024) [17], Batlle-Roca et al. (2024) [18] and Desblancs et al. (2024) [49]. These topics have significant value within today's music industry and the music information retrieval (MIR) research community.

In order to address the impacts of AI, there is a broad corpus of research around the topics of *fairness, accountability* and *transparency*, represented for instance by the FAccT Community[1]. In the EU, the *High-Level Expert Group on Artificial Intelligence*[2] [7] defined the concept of *Trustworthy AI* as a set of seven requirements that an AI system should fulfil along its life cycle: (1) human agency and oversight, (2) technical robustness and safety, (3) privacy and data governance, (4) transparency, (5) diversity, non-discrimination and fairness, (6) environmental and societal well-being and (7) accountability. Transparency—the fourth requirement of Trustworthy AI—is linked to traceability, explainability and communication and it encompasses three main elements of an AI system: data, algorithm and business model [7]. Therefore, AI transparency is the property working towards understanding, describing and interpreting algorithms' characteristics, behaviours and decisions, concerning the receiving audience. Additionally, it considers the needs of different stakeholders, ranging from the developer (i.e. of an AI system, software or service) to the person interacting with the AI system. These principles are critical in addressing the challenges unique to music-generative AI, particularly where questions of originality, copyright, and ethical considerations come into play.

Our study specifically examines transparency in music-generative AI, focusing on models that are primarily based on deep neural networks (DNNs), such as Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), transformers, and diffusion models [15]. While we acknowledge that some generative models may not require training data or follow traditional machine learning (ML) paradigms, our review aims to address transparency concerns within ML-driven generative models. We explore methodologies for assessing music-generative algorithms in terms of creativity and originality, which are critical characteristics in the music field for AI-generated music. We also consider the implications for intellectual property rights, with special attention to music copyright and its link to originality.

To thoroughly analyse the intersection of AI transparency and music-generative AI, we conduct a Systematic Literature Review (SLR), addressing the connections between transparency, artificial intelligence, music generation, the evaluation in terms of creativity and originality, and the links to intellectual property rights. This approach allows

[1] https://facctconference.org/
[2] https://digital-strategy.ec.europa.eu/en/policies/expert-group-ai

us to identify and single out research trends, gaps, and challenges within a defined field [61]. Following a structured review protocol, the SLR ensures a comprehensive and reproducible review process based on defined research questions [102]. This methodology is well-suited for our study as it enables us to evaluate and integrate findings across diverse publications, resulting in an extensive mapping of related literature in AI transparency, creativity, originality and music generation.

However, one of the primary limitations of systematic literature reviews is the time required to complete all necessary steps. This constraint is particularly significant in rapidly evolving fields, such as transparency in generative AI for music. Given the fast pace of publication on this topic, our initial systematic literature review, which included literature published up to December 2022, risked becoming outdated by the time of reporting and sharing. To address this, we employed a two-stage review methodology. The first stage followed the traditional PRISMA approach, focusing broadly on transparency in generative AI and encompassing publications until the end of 2022, as there were then few studies specifically targeting music applications. In a second, focused expansion, we refined our search to include only music-specific publications, capturing emerging trends in music-generative AI. This expansion covered works published from 2023 up to June 2024. Across these two stages, our systematic review comprises a total of 107 selected publications, enhancing its relevance considering the many significant new releases during this period.

With this review, we aim to provide a valuable overview and compilation of relevant literature on transparency in music-generative AI for researchers interested in this particular area, especially for those new to the field. Our key contribution is the identification of research gaps and challenges in this emerging field. In addition, we introduce a dedicated terminology section to clarify essential concepts related to AI transparency (Section 2). Then, the article is structured as follows: Section 3 describes the review methodology; Section 4 presents the findings, addresses the research questions and notes limitations; Section 5 discusses the results and highlights research gaps and challenges of transparency in music-generative AI; and, finally, Section 6 summarises the study's insights and contributions.

2 TERMINOLOGY

An increasing body of literature acknowledges a substantial challenge of terminolgy among AI transparency publications. Most works reveal confusion in the field regarding vocabulary definitions (e.g., Barredo-Arrieta et al. [13], Clinciu and Hastie (2019) [41], Linardatos et al. [85]). In some publications, the terms *explainability* and *interpretability* are used interchangeably, whereas in others their differences are highlighted. Often, this might produce misunderstandings and misinterpretations between researchers. In general, AI transparency terminology tends to encompass closely related and complementary concepts, rather than entirely independent ones. Given the complex nature of generative AI in the music domain, confusion in terminology may affect how principal implication of theses systems are considered and evaluated, making essential to establish clear definitions of AI transparency. For instance, in a music generation model, *explainability* might involve detailing how the system generates a melody from a text prompt to an end-user, while *interpretability* would relate to understanding how variations in input parameters (e.g., music genre or tempo) influence the generated output.

Therefore, we introduce baseline definitions that set the perspective in our review. We define the concept of AI transparency and outline the main methodologies it encompasses. Most of the definitions we introduce were accessed through the *Glossary of human-centric artificial intelligence* [54]: a collection of 230 terms, derived from a variety of sources with the aim to establish a common vocabulary for Trustworthy AI from a human-centric mindset.

- **AI transparency** (based on Trustworthy AI guidelines [7]): Property of working towards unmasking the black-box character of algorithms by understanding and describing their behaviour and interpreting and justifying their decisions and outcomes concerning the receiving audience. It acts as an umbrella for multiple strategies and methodologies, incorporating explainability, interpretability, documentation, reproducibility, auditability and traceability.
- **Explainability** (adapted from definitions in [75] and [13]): The ability to describe the technical processes of an AI system and the reasoning behind its outputs, by providing detailed explanations that allow humans to understand and trace how the system arrives at a particular decision or output. These explanations should be adapted to different levels of expertise, from developers to end users, guaranteeing that the information is understandable across diverse levels of technical knowledge.
- **Interpretability** (combined definitions in [8] and [85]): The capability to comprehend the structure of an AI system and the relation between its inputs and outputs. An AI system is considered interpretable when its behaviour and decisions can be understood by a human.
- **Documentation** (adapted from [74]): To provide data sources origin, structure of the methodology, development procedure and assessment of the results.
- **Reproducibility** (extracted from [7]): "Whether an AI experiment exhibits the same behaviour when repeated under the same conditions".
- **Traceability** (extracted from [7]): "The capability to keep track of a system's data, development and deployment processes, typically through documented recorded identification".
- **Auditability** (extracted from [7]): "The ability of an AI system to undergo the assessment of the system's algorithms, data and design processes. This does not necessarily imply that information about business models and Intellectual Property related to the AI system must always be openly available. Ensuring traceability and logging mechanisms from the early design phase of the AI system can help enable the system's audibility".

3 METHODOLOGY

The present study assesses transparency methodologies and approaches for music-generative AI through a systematic literature review, aiming to identify existing frameworks, highlight research gaps and underscore key challenges in this field. Within this section, we carefully detail each step undertaken to conduct this systematic literature review and outline the research protocol.

Systematic literature reviews provide extensive overviews of large volumes of publications, making them well-suited for addressing specific research questions and identifying research challenges [61]. These reviews follow a predefined and explicit methodology, ensuring reliable and reproducible insights within a specific research field. However, they also come with certain limitations. For instance, they can be highly time-consuming, as completing all phases of the review process may require a considerable amount of time. In addition, using predefined keywords may introduce a bias toward a specific subset of the literature, potentially narrowing the scope and excluding relevant publications. We address the particular limitations of our review in Section 4.6.

Given the rapid evolution of generative AI in the music domain, we implemented a **two-stage** approach to address these challenges while maintaining a systematic approach. First, we included publications up until December 2022, defining this period as our primary search phase (Section 3.2). Acknowledging the growing trend in music-generative AI from 2023 onwards, and to minimize the temporal gap between the search and reporting phases, we conducted an additional search focusing only on music-specific publications from 2023 and 2024. This expansion keeps our review up-to-date by including the latest advancements in transparency for music-generative AI. How we identified these later-added publications is detailed in Section 3.3.

Figure 1 depicts a modified version of the PRISMA 2020 [102] flow diagram, which has been adjusted to reflect the publication screening process for this systematic literature review, including the second-stage focus on 2023 and 2024 publications.

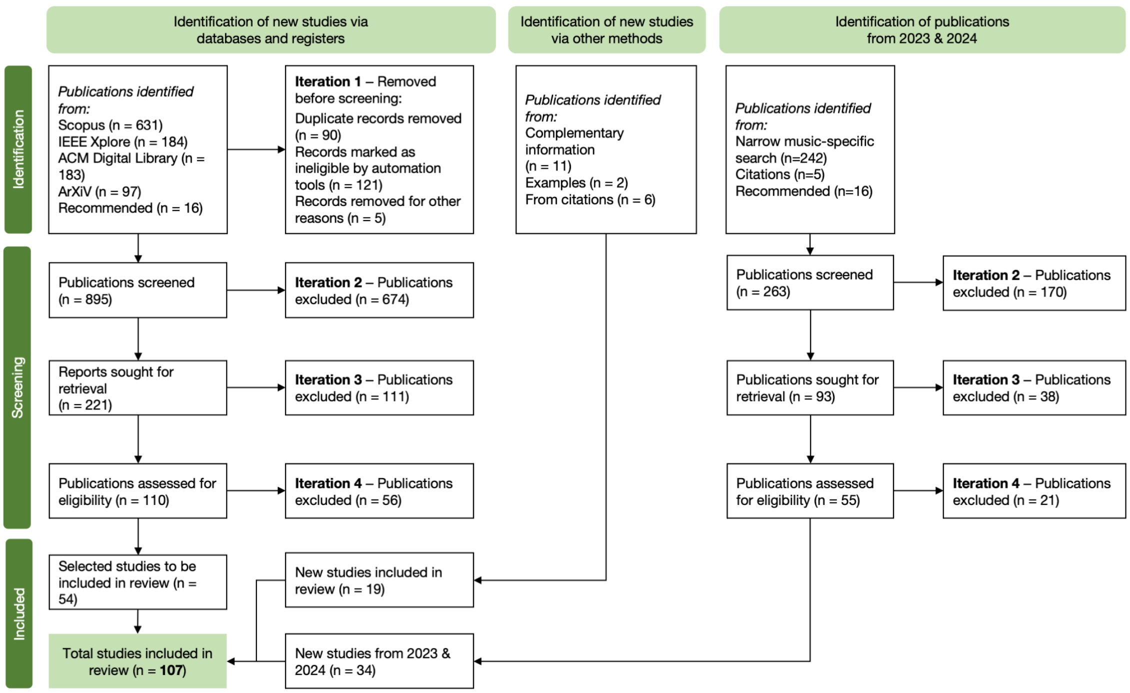

Fig. 1. Modified PRISMA 2020 flow diagram, adjusted to represent the screening process for the SLR, including the extension to 2024.

3.1 Research questions

In a novel area such as transparency in music-generative AI, a systematic approach enables a comprehensive mapping of related literature both in AI transparency and music generation fields. To identify research gaps integrated into all the areas we want to explore, we define specific and precise research questions (RQs). Rather than defining a narrow and direct question that would summarise the target scope of our review, we define four research questions addressing our different areas of interest:

RQ*$_1$*: *What are the current* ***AI transparency methodologies*** *for generative models in the music domain?* We target publications focused on AI transparency strategies, such as explainability, interpretability and documentation, that apply to music-generative algorithms.

RQ*$_2$*: *What are the impacts of music-generative AI on* ***creativity*** *and* ***originality*** *(and* ***intellectual property****)?* Considering the disruption and discussion caused by AI in the digital art domain, we intend to observe the consequences and challenges of generative AI in music on creativity and originality and, subsequently, the derived implications on intellectual property rights.

RQ*$_3$*: How can **creativity and originality** in AI-generated music be **evaluated and measured? Creativity and originality are two controversial concepts with many investigations regarding how they could be evaluated both objectively and subjectively. We aim to map current evaluation strategies and their fit for assessing the creativity and originality of AI-generated music.

***RQ*$_4$*: What is the **role of transparency** in the assessment of creativity and originality of AI-generated music?* With this last research question, we intend to overview the implementation of transparency specifically for the music generation domain, with special interest in its contributions to the assessment and evaluation of AI-generated music creativity and originality, and the potential implications to intellectual property rights.

3.2 First-stage: PRISMA methodology

3.2.1 PRISMA 2020 framework. To conduct a thorough systematic literature review, various frameworks are available. One of the most widely used is PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses). In the past years, different versions of this methodology have been proposed, e.g., [95, 102, 109]. For our investigation, we implemented *PRISMA 2020* strategy [102], which is conducted in three phases:

(1) **Planning**: to identify the need for the review, specify the research questions and define a review protocol.
(2) **Conducting**: to determine the principal search sources, select the primary publications, perform a quality assessment, extract and monitor the relevant information and synthesise the data.
(3) **Reporting**: to integrate, format and evaluate the information and report.

3.2.2 Search strategy. The search strategy consists of all the steps followed to identify and select suitable papers to cover the set RQs. For our investigation, we looked for articles in four popular publications databases: Scopus[3], IEEE Xplore[4], ACM Digital Library[5] and arXiv[6]. Among these, Scopus, IEEE Xplore, and the ACM Digital Library provide access to peer-reviewed publications, while arXiv hosts preprints and non-peer-reviewed works. We included a non-peer-reviewed database to avoid missing out on novel-related publications resulting from the rapid transformation in the music generation field. In addition, we consider additional publications suggested by authors and colleagues.

For each proposed research question, we defined a group of suitable keywords, tailored to the specific focus of each question. This differentiation of keyword selection was necessary to ensure a diverse range of literature was obtained and provide a more comprehensible analysis. Each keyword belongs to a *field*, which corresponds to the synonym category to combine keywords. For example, keywords *explainability*, *interpretability*, and *documentation* belong together to the field *transparency*. How these keywords are associated depends on each question and the relevance of principal keywords, which served as a search constraint[7]. Not using a search constraint resulted in an ineffective search as an unmanageable amount of publications were retrieved for RQ$_1$ and RQ$_2$ (e.g., over 60.000 detected entries in Scopus). Table 1 exemplifies the keyword combination applied in each research question. In the search queries, we used the roots of the words to include all possible derivatives, for example: *original** would include *original, originality* or *explainabl** would include *explainable, explainability.* Our use of modern terminology is intentional, as it aligns with the field of AI transparency. However, we acknowledge that focusing our search towards this modern terminology may limit the retrieval of earlier publications that used different keywords or vocabulary. While we recognize the value of historical

[3]https://www.scopus.com/
[4]https://ieeexplore.ieee.org/Xplore/home.jsp
[5]https://dl.acm.org/
[6]https://arxiv.org/
[7]A search constraint refers to a specific requirement applied in database searches, where a particular keyword must appear in a designated part of the publication, such as the title, abstract, or list of keywords. This technique helps to refine search results by ensuring that relevant articles are retrieved.

perspectives, the rapid advancements in AI and music generation motivate a focus on the most recent research to capture the state-of-the-art developments in this domain.

Despite our questions targeting the music field, we included other relevant areas considered to be more advanced in transparency for generative algorithms (e.g., image, speech, text). This decision was made to benefit from their findings and contributions. Besides, considering the novelty of Trustworthy AI and AI transparency, we decided not to have any time of publication limitation. Thus, all the selected articles were published online by December 2022, when the search phase was conducted.

In our search, a publication is identified as suitable for an RQ, if it appeared in any of the search engines when the keyword combination complied with the title, abstract and defined keywords. In the case of arXiv, we only considered titles as a large number of publications were detected. Adapting search queries to each search engine is not an uncommon practice in systematic literature reviews, as we can observe in other works (e.g., Ciancarini et al. [37]).

Table 1. Search query per each RQ. In bold, the keywords used as search constraints.

RQ	Search query
1	**(transparency OR transparent-artificial-intelligence)** AND (audio OR image OR language OR music OR music-creation OR music-generation OR speech OR text) AND (algorithm OR artificial-intelligence OR generative-model OR machine-learning OR generation-algorithm) AND (auditability OR black-box OR comprehensibility OR controllability OR documentation OR explainability OR explainable-artificial-intelligence OR XAI OR interpretability OR reproducibility OR traceability OR trust OR trustworthy artificial intelligence OR uncertainty OR understandability OR end-user OR human-in-the-loop)
2	**(creativity OR intellectual-property OR authenticity OR copyright OR originality OR regulation)** AND (audio OR image OR language OR music OR music-creation OR music-generation OR speech OR text) AND (algorithm OR artificial-intelligence OR generative-model OR machine-learning OR generation-algorithm)
3	(originality OR authenticity) AND (music OR art OR poetry OR painting) AND (evaluation OR measure OR assess)
4	(transparency OR transparent-artificial-intelligence) AND (originality OR creativity OR intellectual-property OR copyright)

3.2.3 *Inclusion and exclusion criteria.*

Systematic literature reviews tend to be extensive and compile a large list of potentially corresponding publications. Hence, it is essential to define strong inclusion and exclusion criteria and be consistent throughout the publication selection process. Therefore, we included only publications that fulfilled the subsequent criteria and excluded others that did not:

- The work addresses or is related to one or more research questions.
- The work contributes to up-to-date discussion and brings relevant insights to the investigation.
- The work has to be published or available online by December 2022.
- The work is available in English, is not duplicated and is complete.

3.2.4 *Publication screening.*

An initial search, including the recommended publications, provided 1,111 suitable publications. Table 2 shows the number of publications retrieved from each search engine and research question. The appraisal or publication screening phase consisted of four iterations. In each iteration, a publication was accepted into the next step if its content was enhancing and contributing to our understanding of the topic and target research questions:

Iteration 1	Remove duplicates, non-complete works and not-related topics.
Iteration 2	Review the title, keywords and abstract.
Iteration 3	Review the title, abstract, introduction, conclusion, figures and tables.
Iteration 4	Read and examine carefully the selected publications

Table 2. Amount of publications identified per search engine and research question for the first stage of the review.

Search Engine	RQ_1	RQ_2	RQ_3	RQ_4	Total Publications
Scopus	159	219	116	137	631
IEEE	45	70	36	33	184
ACM	43	132	6	2	183
arXiv	47	7	36	7	97
Suggested	5	3	2	6	16
	299	431	196	185	1,111

During the first iteration, we detected 90 duplicates and 5 non-complete works. We removed off-topic publications: image watermarking (95) and cryptography (26). At the end of this iteration, we reduced the number of publications to 895, 13 of which belonged to more than one question (RQ_1=238, RQ_2=381, RQ_3=169, RQ_4=94). Next, the second iteration allowed us to have a clearer perspective on the content of the publication by reviewing the title, keywords and abstract. We discarded a large number of unrelated or not significantly relevant publications, reducing the identified publication list to 221 publications, of which 13 were shared by more than one question (RQ_1=98, RQ_2=78, RQ_3=20, RQ_4=39). The third iteration provided a more detailed perspective of the analysed publications. Especially when looking at the introduction, contributions and future work. This procedure allowed us to determine which from the analysed publications were more aligned with our investigation. From this iteration, we obtained 110 potential publications, of which six were shared by more than one question (RQ_1=34, RQ_2=54, RQ_3=12, RQ_4=16). Finally, the fourth iteration enabled us to get a viewpoint of the explored field, understand what strategies are being implemented and their outcomes, and consider their implications and characteristics. This procedure acted as a major quality assessment of publications as we carefully examined each of them. We discarded publications when they did not contribute to our review or if their contribution was out of the scope of this review. For instance, during this iteration, the book *The Creative Mind: Myths and Mechanisms* [21] was updated to its second edition.

We obtained 54 suitable publications, 3 of which are shared by two research questions (RQ_1=20, RQ_2=22, RQ_3=9, RQ_4=6). There is an unbalanced number of publications per research question, due to the specificity and novelty of each question. For example, AI transparency (RQ_1) is a more studied topic, especially taking into account other fields, than determining whether AI-generated music is original (RQ_3) or contributions of transparency applied to music-generative AI (RQ_4). Nonetheless, during the information extraction phase, we identified complementary publications through citations (6). We complemented our study with specific use cases and examples related to the assessed topics, as well as additional information to provide a more comprehensive answer to our RQs (13). We made this decision because, for certain topics, such as legal regulation of generative art, we retrieved a few related publications in the appraisal phase. In the end, we identified **73** suitable publications (RQ_1=27, RQ_2=30, RQ_3=11 and RQ_4=8), 3 of which belong to two research questions.

3.3 Second-stage: Music-focused expansion

3.3.1 Motivation. The rise of AI in digital art creation is advancing at a fast pace, with novel generation models in the audio-visual domain, mainly driven by a strong push from the industry. In 2023 and 2024, we witnessed the popularisation of music generation models with promising contributions such as *MusicLM* [5], *Moûsai* [117], *MusicGen* [44], *SingSong* [50], *MusicLDM* [35], *VampNet* [60], *Stable Audio* [56], *Stable Audio Open* [57], *Suno*[8] and Udio[9]. These developments underscore the increasing relevance and rapid growth of the music generation field. Moreover, there has been a substantial raise of workshop related to AI transparency in arts and audio generative domains, such as the workshop on *Artificial Intelligence and Creativity (CREAI)*[10] since 2021, the workshop on *EXplainable AI for the Arts (XAIxArts)*[11] since 2023 or the workshop on *Explainable AI for Speech and Audio (XAI-SA)*[12] at ICASSP 2024. While these events are not music-specific, they exemplify the growing trend on AI transparency in the digital-art and audio generation domain, which is key for encouraging research on transparent generative AI systems in the music domain.

One of the principal limitations of systematic literature reviews is the constraint of time. Completing all steps in a systematic review process requires a significant amount of time, a notable challenge particularly important in rapidly evolving fields such as music-generative AI. To minimize the temporal gap between the search phase of the review (which was conducted in December 2022) and the dissemination of the results, we conducted a narrow music-focused extension of the identified publications to include publications from 2023 up to June 2024 (included). This second-stage review was crucial as 2023 and 2024 have witnessed an explosion of new music-generative AI models and significant developments in transparency methodologies, which would otherwise be omitted from our analysis.

3.3.2 Search strategy and publication screening. We employed a music-specific approach to prioritise developments in music-related generative AI while managing the scope to fit within our research timeline. We searched for relevant publications in Scopus and arXiv using the previously defined keywords for each research question. We included additional search constraints to limit the search to the music domain and generative AI: "music OR music generation OR music-generative AI OR generative AI OR music creation".

Our updated search yielded 253 potentially suitable publications and, in addition, we included 10 publications suggested by authors and colleagues. Consequently, this expansion up to June 2024 consists of 263 potentially suitable publications. To screen this new set of publications, we followed the same screening procedure as for the first stage. However, we skipped Iteration 1 (see Figure 1). We considered 93 of them to fit our review, based on reviewing the title, abstract and keywords. After carefully reviewing and appraising the identified publications (Iteration 4), we determined that **34** of them were particularly relevant to our investigation (RQ_1=5, RQ_2=13, RQ_3=10, RQ_4=15—9 of which are shared by two questions). The significant findings of these publications are integrated into the results of the RQs presented and discussed in the subsequent sections.

3.4 Data extraction and analysis

For each identified publication in the search phase, we registered the essential information of the work: title, authors, abstract, year, editor/publisher, keywords, paper type, URL or DOI (Digital Object Identifier) and number of citations. We indicated the research question it belonged to and the process used to identify the publication (i.e., keyword combination,

[8]https://suno.com/
[9]https://www.udio.com/
[10]CREAI 2024: https://creai.github.io/creai2024/
[11]XAIxArts 2024: https://xaixarts.github.io/2024.html
[12]XAI-SA 2024: https://xai-sa-workshop.github.io/web/

suggested publication). In each iteration of the appraisal phase of this SLR, we specified the perceived contribution that each publication had to our investigation in a 3-level scale rate: 1=**not impactful**—should be discarded, 2=**somewhat impactful**—should be considered to be included, 3=**impactful**—should be included. Only works classified in the third level (**impactful**) were reviewed in the subsequent iteration. We identified the most relevant aspects of the work that were a beneficial contribution to our search (i.e., methodology, strategy, discussion, proof of concept, research motivation).

While conducting the fourth step of the appraisal phase (i.e., carefully reading the selected publications), we summarised the contributions of the selected articles by analysing the data and comparing it with the other publications. As previously mentioned, when extracting and analysing the information from the identified publications, we expanded our identified publications by incorporating supplementary references through citations and additional sources to fulfil the scope of our research questions. This decision was motivated by the will to add information on specific topics where we faced limitations in retrieving enough contributing publications. The outcomes and discussion of the data extraction and analysis are detailed in the upcoming sections (Section 4 and 5).

4 RESULTS

4.1 Identified publications

The appraisal phase of this systematic literature review started with 1,111 potentially suitable publications. Following the PRISMA review process outlined earlier, we identified 72 publications. Later, we expanded the review with 34 publications from 2023 and 2024, bringing the total to **107** publications. All identified publications are detailed in Table 3, where they are classified based on their main subject of contribution, which was determined during the fourth phase of the SLR—involving a careful review of the publication and extraction of the relevant data. This classification was intentionally designed to align more closely with the specific content of the publications, allowing for a more effective grouping and aiming to provide a clearer summary for readers engaging with the review. In addition, we provide a focused summary of transparency-enhancing strategies in Table 4. This table presents the 10 most representative publications of the review, which present approaches to enhance transparency within music generative models.

The extracted information has been used primarily to answer the target RQ. However, some publications overlap with each other or contribute to more than one research question. Consequently, in some instances, we used information from one RQ to complement our findings in another. This decision was made because certain insights were relevant to multiple questions and could provide valuable perspectives for different aspects of our research. This particularly happened when answering RQ_3, which was complemented with publications identified in RQ_2.

Figure 2 depicts the most frequently occurring words in the titles of the selected publications, excluding terms *artificial intelligence* and *music*. Notable terms include *creativity*, *generative/generation*, *model* and *art*, which highlight the central themes of creative processes and generative approaches within the field. Additional prominent terms such as *explainable*, *evaluation*, *interpretability*, and *human* indicate a strong focus on transparency, assessment, and the human aspect in generative AI research. The recurrence of terms like *explainability* and *interpretability*—core elements of AI transparency—aligns with the review's structured search strategy, which prioritizes recent advancements in understanding and assessing AI's transparency within creative domains, particularly in music. However, we must acknowledge that our search strategy is biased towards more modern times due to the novelty of the AI transparency field and the keywords used linked to recently introduced vocabulary.

Table 3. Identified publications

AI Transparency	Abdollahi and Nasraoui [1], Rudin [112]
Explainability	Adadi and Berrada [2], Afchar et al. (2022) [3], Arias-Duart et al. [10], Barredo-Arrieta et al. [13], Bryan-Kinns et al. (2021) [29], Bryan-Kinns et al. (2024) [30], Clinciu and Hastie [41], Clinciu et al. (2021) [40], Ehsan et al. [51], Henin et al. [68], Miller [92], Mishra et al. [93], Noel-Hirst and Bryan-Kinns [101], Pintelas et al. [106], Ribeiro et al. [110], Samek et al. [113], Wang et al. [127]
Interpretability	Alonso-Jiménez et al. [9], Böhle and Fritz [24], Linardatos et al. [85], Molnar [96], Schröder and Ghajagar [118], Sokol and Flach [120], Zinemanas et al. [137]
Documentation	Arnold et al. [11], Henri et al. [62], Holland et al. [72], Hupont et al. [74], Mitchell et al. [94], Morreale et al. [98]
Music Generation	Agostinelli et al. [5], Briot and Pachet [28], Chen et al. [35], Ji et al. [76], Ragot et al. [108], Simões et al. [119], Zhu et al. [136]
Creativity & Computational Creativity	Arriagada [12], Bian et al. [19], Birtchnell and Elliot [20], Boden (2004) [21], Boden (2010) [22], Boden and Edmons [23], Caramiaux and Alaoui [31], Carnovalini and Rodà [32], Casini et al. [33], Cetnic and She [34], Chung [36], Cotton and Tatar [46], Daniele and Song [48], Epstein et al. [52], Esling and Davis [53], Ford et al. [59], Hitsuwari et al. [71], Jordanous (2010) [77], Kilb and Ellis [81], Loughran and O'Neill [86], Louie et al. [87], Natale et al. [100], Pasqueier et al. [104], Rohmeier [111], Sato and McKinney [116]
Creativity Evaluation	Agres et al. [6], Cádiz et al. [47], Colombo et al. [42], Guo et al. [65], Horzyk [73], Jordanous (2011) [78], Jordanous (2012) [79], Lamb et al. [83], Mejtoft et al. [91], Moruzzi [99], Xiong et al. [131], Yin et al. (2023) [135]
Co-Creativity	Correia et al. [45], Hirawata and Otani [70], Kamath et al. [80], Oever et al. [126], Qiu et al. [107]
Originality and Plagiarism in Generation Models	Afchar et al. (2024) [4], Barnett et al. (2024) [17], López-García et al. [88], Somepalli et al. [121], Yang and Lerch [132], Yin et al. (2021) [133], Yin et al. (2022) [134]
IP and Copyright	Bonadio and McDonagh [25], Hayes [67], Lee et al. [84], Mantenga [89], Mazzi [90], Strum et al. [123], Sueur et al. [124]
Law Specific	UK Intellectual Property Office [63], US Copyright Office [64], Samuelson (2023) [114], Samuelson (2024) [115]
Generative AI Implications	Bandi et al. [15], Barnett (2023) [16], Clancy (2021) [38], Clancy (2023) [39], Knott et al. [82], Morreale (2021) [97], Tang et al. [125]
Miscellaneous	Baeza-Yates and Estévez-Almenzar [14]

When examining the year of publication of these investigations (Figure 3[13]), a clear trend of increasing interest in AI transparency emerges: 95% of the selected works have been published after 2016. For instance, both Adadi et al. [2] and Linardatos et al. [85] point out a growing trend of Explainable AI (XAI) from 2016, indicated by the Google Trends Popularity Index[14]. With the same tool, we illustrate in Figure 4 popularity trends for *music generation*, *AI transparency*, *explainability*, *interpretability*, *AI creativity*, and *AI originality* in the last five years. *Music generation* popularity comes before 2020, with some rising points in 2022 and 2024. *AI creativity* and *interpretability* show some fluctuations, suggesting variable interests, potentially influenced by advancements in research and media coverage.

[13]Note that two of the identified publications [39, 71] were available online in the last months of 2022, but their official publication was in early 2023. Consequently, they appeared in our SLR search phase, and are classified as 2022 publications, but their citation year corresponds to 2023.
[14]https://trends.google.com/trends/

Table 4. Most relevant publications that enhance transparency in music generation models.

Reference	Year	Venue or Publisher	Main Contribution or Highlight
Afchar et al. (2024) [4]	2024	arXiv	Introduce a music **deepfake detector** with 99.8% accuracy, highlighting challenges in calibration, robustness, and interpretability for future research.
Barnett et al. (2024) [17]	2024	ISMIR	Develop a **method to identify training data influence** using audio similarity, validated by human listening studies to aid transparent model output attribution.
Bryan-Kinns et al. (2021) [29]	2021	XAI4Debugging (NeurIPS)	Extends MeasureVAE with **XAI features**, enabling real-time control and visualization of musical attributes in the latent space for improved transparency and user interaction.
Bryan-Kinns et al. (2024) [30]	2024	Machine Intelligence Research	Compares measureVAE and adversarialVAE, finding that measureVAE performs better in **music generation with interpretable dimensions**, especially for low-complexity genres.
Chen et al. [35]	2024	ICASSP	Introduces **MusicLDM**, a text-to-music diffusion model, **addressing plagiarism risk** by adding mix-up strategies to reduce memorization, verified through objective metrics and listening tests.
Guo et al. [65]	2023	PLOS One Journal	Proposes MEM, an **evaluation method** combining statistics, theory, and signal processing for enhanced model transparency.
Noel-Hirst and Bryan-Kinns [101]	2023	XAIxArts	Autoethnographic study of the MeasureVAE XAI model, demonstrating its potential to **enhance music-making workflows**.
Yin et al. (2021) [133]	2021	EvoMUSART 2021	Introduce the **originality report** to assess replication in music models, finding low originality and overfitting issues in the Transformer model.
Yin et al. (2022) [134]	2022	SN Computer Science	Expands the originality report with a **similarity score**, confirming Transformer's copying issues persist even with larger datasets.
Xiong et al. [131]	2023	arXiv	Comprehensive **survey** of **evaluation** methodologies for AI-generated music.

From 2023 onward, we can observe a clear rising trend for all terms, especially for *AI originality*. As shown in Table 5, in more recent years, there has been a notable increase in publications related to RQ3 and RQ4, reinforcing the correlation between rising interest in AI transparency and the number of literature contributions in these areas. This data indicates that researchers are increasingly exploring the intersection of transparency in AI and music generation, reflecting both a growing academic focus and a response to emerging ethical considerations in the music domain.

4.2 RQ_1: What are the current AI transparency methodologies for generative models in the music domain?

Our first research question, RQ_1, is focused on reviewing AI transparency methodologies for generative models in the music domain. We found very few specific investigations addressing AI transparency in the context of music generation throughout our systematic literature review. Since our search strategy combined other domains in the search

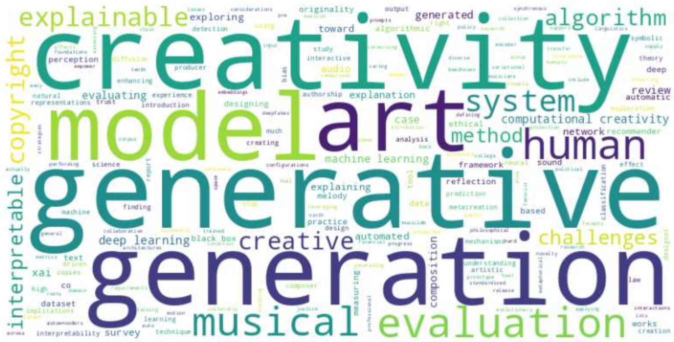

Fig. 2. Word cloud of the titles of the 107 selected publications, without 'AI' and 'music'.

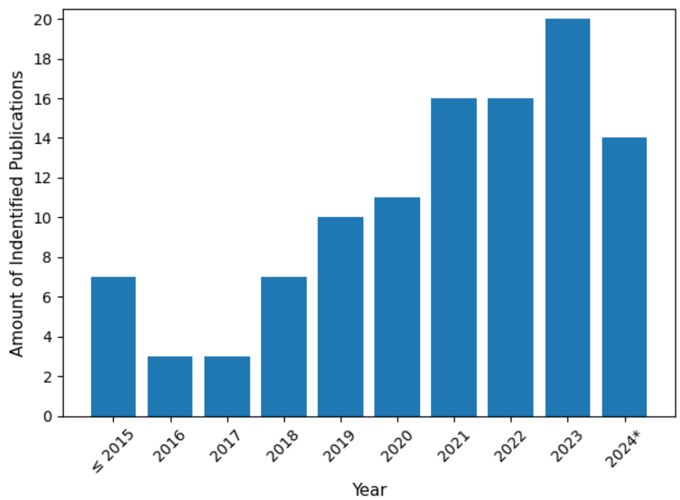

Fig. 3. Number of identified publications per year beyond 2016.

*Note that for the year 2024, only publications until June (included) have been selected for the review.

query (i.e. text, image, speech), we obtained a comprehensive overview of state-of-the-art transparency methodologies with a general and diverse AI perspective. The identified literature led us to review three key areas to enhance transparency: explainability, interpretability and documentation. We summarise the retrieved results in the next subsections, considering general and specific approaches that could be applied to generative AI in the music domain. Thus, this section acts as an introduction to AI transparency methodology, which is beneficial for readers without previous knowledge in the field.

4.2.1 Explainability. One of the most popular methodologies comprised within AI transparency is explainability (alternatively referred to as Explainable AI or XAI). Explainable AI aims to propose approaches and methodologies that yield more understandable and trustworthy models by describing the technical processes of an AI system, relating them to human decisions, and to provide explanations that are adapted to various levels of expertise, ensuring that

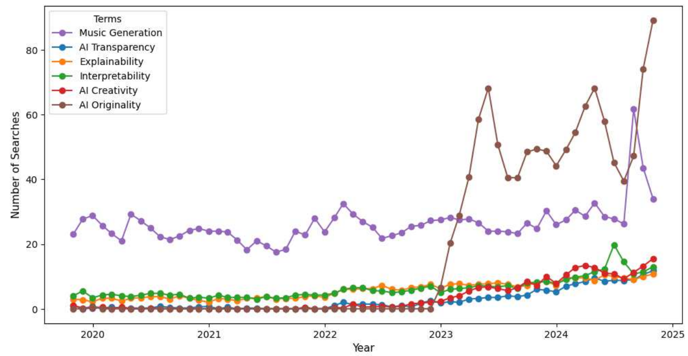

Fig. 4. Monthly popularity trends of key terms of this review from October 2019 to October 2024.

Table 5. Publication counts per year and research question (RQ).

Year	RQ1	RQ2	RQ3	RQ4	Total**
≤ 2015	0	5	2	0	7
2016	1	1	1	0	3
2017	1	1	1	0	3
2018	3	4	0	0	7
2019	7	2	1	1	10
2020	2	6	2	1	11
2021	9	4	2	3	16
2022	4	7	2	3	16
2023	2	8	6	8	20
2024*	3	5	4	7	14
Total	**32**	**43**	**21**	**19**	**107**

*Note that for the year 2024, only publications until June (included) have been selected for the review.

**Note that 12 publications belong to two research questions.

the information is comprehensible to both developers and end users (see Section 2). These strategies can contribute to detecting biases and quality issues beyond describing the inner process of an algorithm [1, 113]. In the literature, we find multiple approaches for explainability in ML. However, we only identify explainable AI methodologies directly implemented in music generation algorithms in recent publications.

We identify two deep and detailed surveys on XAI: Barredo-Arrieta et al. [13] and Adadi et al. [2]. Noted in Barredo-Arrieta et al. [13], explaibility approaches are divided into two categories: (1) *transparent models*, which are interpretable by design, and (2) *post-hoc explainability models*, which include external explainability techniques and can be model-agnostic or model-specific. In our review, we found more initiatives focused on post-hoc explanations.

Regarding post-hoc explanations, Samek et al. [113] provide a complete overview focused on DNNs and point out the main challenges and limitations. Nowadays, one of the most significant and deployed post-hoc explanation techniques in the XAI community is LIME (Local Interpretable Model-agnostic Explanations) [110]. It is characterised by being a novel approach to explaining models' predictions by providing local explanations of any classifier. Later on, Mishra et al. [93] extend the application of LIME to music content analysis. They propose three versions of explanations based on temporal segmentation, and frequency and time-frequency segmentation, which allow us to understand the influencing factors on input data classification. Their investigation reveals insights into model behaviour and demonstrates that the decision tree model struggles with generalization. This demonstrates the possibility of implementing general or other domain-specific strategies in the music domain.

However, explanations may have multiple content and representations, depending on their intended recipients. This is why many of the identified publications in this review emphasise the relevance of focusing and tailoring explanations to a specific end-user[15] [1–3, 13]. It is essential to centre the explanations towards the corresponding audience profile and state why the explanation is requested [13]. For example, in music recommender systems, to understand why a song was suggested in a playlist, listeners and developers require different information. At the same time, it is critical to consider the role to whom the explanation is targeted. Providing too much information can overwhelm the audience with irrelevant information for them [3]. That is, it is better to provide accurate and relevant explanations, rather than all the available information. Towards this end, a framework for black-box explanations proposed by Henin et al. [68] can be employed by a wide range of audiences, including non-specialized users, and adapts to multiple types of explanations.

Nonetheless, most of the identified publications point out the need to evaluate the performance of the explanations. The lack of formality and clear definition on an explanation task, does not allow for comparing performance between strategies nor guaranteeing that the explanations are understandable by the set audience [3, 40]. Towards solving this issue, Clinciu et al. (2021) [40] propose an evaluation strategy to analyse how effective explanations are, taking into account the user's understanding. Furthermore, Miller [92] highlights the importance of going beyond explicitly explaining decisions or actions to the target audience. To develop *good* explanations, researchers should benefit from the vast literature on cognitive and social psychology on human reasoning and understanding. In line with this perspective, Wang et al. [127] propose a theoretical framework based on decision theory to help developers build more human-centric explanations. With a similar perspective, Arias-Duart et al. [10] acknowledge the lack of assessment of explanations in the image domain. Consequently, they propose *Focus*: a reliable metric to evaluate the precision of feature attribution methods in the image recognition field. This metric can be re-used to assess the reliability of explanations and identify and characterise bias.

Focused on music recommender systems, Schröder and Gjakargar [118] explore the benefits of conducting a dialogue between the system and end-users. Their study concludes by describing how this strategy improves algorithm understandability and stressing the need to design more detailed systems to achieve more transparent algorithms for end-users.

We encountered publications intending to contribute with explainable methods for music-generative algorithms from 2021. For instance, Bryan-Kinns et al. (2021) [29] provide a more explainable music generation model based on MeasureVAE [105]. The model's explainability is augmented by regularising the latent space, introducing a user interface feedback loop enabling real-time adjustment of latent space dimensions, and visualizing musical attributes in the latent space for better understanding and prediction of the impact of dimension changes. In addition, this work acknowledges that explainable models focused on generative art are scarcely explored. In subsequent work, Bryan-Kinns et al. (2024) [30] explore two Variational Auto-Encoder (VAE) models, latent space configurations, and training datasets on music-generative AI performance by including semantic features. They intend to make algorithms for music generation more understandable.

[15]Hereafter referred to as *audience*, a term used in the literature to encompass the different profiles to whom an explanation might be directed, including artists, developers, listeners, among other stakeholders (see Barredo-Arrieta et al. [13] for further discussion on this topic).

4.2.2 Interpretability. Focusing on interpretability, Linardatos et al. [85] contribute with an extensive and detailed review of ML interpretable methods. One of their main claims is justifying the difference between *explainability* and *interpretability*. Considering the definitions introduced in Section 2, explainability aims to provide detailed insights into the internal processes and reasoning of a model, allowing users to trace how specific decisions are made. In contrast, interpretability focuses on making the relationship between inputs and outputs more intuitive, ensuring that humans can understand and predict the model's behavior without needuing in-depth technical knowledge. Therefore, these concepts alone are insufficient and must complement each other to enhance models to be more . Moreover, thistransparent investigation acknowledges the multiple characteristics interpretability techniques have and establishes that all of them should be considered, rather than classified into a single category. Consequently, they propose a four-category taxonomy reflecting the different aspects an interpretability method could be classified into:

(1) Purpose of interpretability
(2) Type of algorithm it can be applied to: model-agnostic, model-specific
(3) Scale of interpretation: specific instance (local) or whole method (global)
(4) Type of data

In parallel, many publications state there is a dynamic trade-off between model performance and achieving interpretable predictions. Overcoming this challenge requires interdisciplinary collaboration [2, 113]. Nonetheless, recent literature demonstrates that similar performance can be achieved when using an interpretable model (e.g., [24, 106]). Thus, powerful performance from deep learning algorithms is not incompatible with interpretability.

In the audio classification domain, Zinemanas et al. [137] propose APNet (Audio Prototype Network), a novel interpretable prototype-based model. APNet classifies audio by comparing the similarity of the input sound to learned prototypes in the latent space. In addition, through the use of an autoencoder, the model enables the sonification of prototypes, providing further interpretability in the classifications. Their model achieves comparable performance to non-interpretable state-of-the-art models across three different sound classification tasks: speech, music and environmental sounds. Building on APNet, Alonso-Jiménez et al. [9] introduced PECMAE (Prototype EnCodecMAE), a music genre interpretable classifier that improves upon APNet by training the autoencoder and prototypical network separately. This allows PECMAE to leverage pre-trained self-supervised models, achieving competitive results on the more complex task of music genre classification while maintaining interpretability.

All of these investigations align with the recommendation by Rudin [112] on promoting and building models that are inherently interpretable rather than trying to explain black-box models. However, none of the identified publications specifically targets interpretable approaches for music generation models,

4.2.3 Documentation. Another principal strategy for transparency is documentation. In recent years, novel research and industry-promoted methodologies for documentation have appeared [74]. For instance, *Datasheet for Dataset* [62] is a proposed method to work towards standardization of dataset documentation in the machine learning community. It intends to promote that every dataset is accompanied by a datasheet detailing all its characteristics, aims and sources. With a similar focus, *The Dataset Nutrition Label* [72] introduces a flexible and adaptable framework to assess the quality of datasets used in AI development, to improve performance and reduce bias. This strategy brings many benefits, such as enhancing data practices, facilitating the selection of optimal datasets and increasing the quality of algorithms.

Not all documentation strategies are centred on dataset description. Some investigations also address the documentation of AI models and algorithms. For instance, *Model Cards for Model Reporting* [94] proposes a framework based on short documents that contain benchmark evaluations of the machine learning model. These documents provide information regarding performance, intended use cases, and potential biases. This approach empowers users to determine a model's suitability for their particular use case.

Alternatively, *FactSheets* [11] proposes to increase trust in AI services with a set of documents reporting purpose, performance, safety, security and provenance information, from the perspective of a *supplier's declaration of conformity* (SDoC). Note that this work is distinguished from the previous cases by focusing on the final AI service, rather than an ML model or a dataset.

Notably, none of the identified documentation strategies specifically addresses music-generative models, highlighting a potential area for future research to tailor these methods to the unique needs of AI-driven music systems.

4.3 RQ$_2$: What are the impacts of music-generative AI on creativity and originality (and intellectual property)?

RQ$_2$ aims to explore how music-generative AI influences creativity and originality. In addition, we intend to consider the consequences of AI-generated music for intellectual property rights, including potential issues such as IP infringement and violations.

4.3.1 Creativity. Creativity is one of the most questioned and studied characteristics in human and philosophical studies, filling the literature with multiple definitions. Nonetheless, one of the most accepted definitions and constantly reaffirmed in the literature is the one proposed by creativity philosopher Margaret Boden: "Creativity is the ability to come up with ideas or artefacts that are new, surprising, and valuable" [21].

Computational Creativity (CC) is a multidisciplinary field that studies and contributes to developing creative (or potentially creative) algorithms. One of its most popular subfields, and the target of this survey, is music generation (alternatively known as Musical Metacreation [32]). From the reviewed publications, the most relevant works tackling this issue in the music domain are Carnovalini and Rodà [32] and Cádiz et al. [47]. For instance, Carnovalini and Rodà [32] present an introductory survey on computational creativity and music generation systems state-of-the art. They expose the stated issue of creativity, review three popular approaches to evaluate computational creativity, observe the main approaches to music generation and identify its open challenges. Then, Cádiz et al. [47] introduce a review of computational creativity and music-generative models. Their work points out the need to challenge the understanding and evaluation of computational creativity. Aligned with these two investigations, Briot and Pachet [28] delve into the challenges and limitations of deep learning in music generation, particularly focusing on issues of control, structure, creativity, and interactivity.[16] They point out that these models lack controllability over musical elements, such as tonality and rhythm, leading to outputs that often mimic training data without showing true creativity. This article highlights these critical limitations, emphasising the need for approaches that can make generative systems more transparent and adaptable to human input, which is still a significant challenge for the field.

During our review, we identify two main and opposite perspectives on machine creativity. Other investigations disclose this disagreement too (e.g., [99]):

(1) Machines cannot be creative without human intervention (e.g., [48])
(2) Machines have the capability to be creative (e.g., [12]).

On the one side, Daniele and Song [48] state that "AI as a technical object is always intertwined with human nature despite its level of autonomy". In their work, they emphasise the need to enhance transparency strategies, while being aware of the implications of technology to society and the conception of creativity, and claim this discussion must go

[16]For further details and a more comprehensive analysis refer to Briot et al. (2020) [27].

beyond the CC community. They do not see AI as more than a tool to make art, as a violin to a violinist. From their perspective, there is no AI art without human involvement and participation.

On the other side, Arriagada [12] sustains that believing that AI cannot be creative is due to an anthropocentric bias—that is, to think that humans are the centre of everything. Another present argument here is that AI enriches human art, rather than trying to replace it. Both Louie et al. [87] and Arriagada [12] state the power of AI as a co-creation tool, to help artists and enhance human creativity, rather than a threat to substitute artists. This argument is further supported by Epstein et al. [52], where they acknowledge generative AI's role as a disruptive tool within the arts, but with the capabilities to enhance human creativity. Epstein et al. provide an in-depth analysis of the influences of AI on creative practices, advocating for "meaningful human control" to maintain human influence over AI outputs.

Nowadays, a significant number of works are being carried out to enhance AI as a co-creation tool in music creation processes. For instance, a couple of recently presented collaborative AI-human tools for music composition are *MoMusic* [19] and *Humming2Music* [107]. MoMusic allows users to control harmonic progressions generations in real-time employing an interactive music composition system, based on motion recognition. Instead, Humming2Music enables users to create full compositions simply by humming, including modules that transcribe, generate melodies, and add accompaniment based on the user's input. These tools illustrate novel paths for enhancing and accompanying creativity with AI. Furthermore, Hitsuwari et al. [71] presents a case study on haiku poetry where AI-human collaboration enhances beauty rating, compared with only human or only AI works.

A recent publication by Brandt [26] presents a study comparing human and computational creativity linked to Beethoven's Ninth and unfinished Tenth symphonies. The investigation assesses *how well* a specifically trained generative AI model can suggest a continuation of Beethoven's Tenth. While the generative model successfully mimics aspects of spontaneous human creativity, the results show it falls short in capturing deliberate creativity—which involves non-linear thinking, context-driven decision-making, and iterative revision—, which was notably absent in the AI's approach. This publication underscores the importance of multidisciplinary collaboration among musicians, cognitive scientists, and computer scientists, revealing AI's strengths and limitations in creative problem-solving.

4.3.2 Originality (and intellectual property). A generation algorithm is expected to balance between producing similar outputs and not replicating the training data [47]. This expectation raises critical questions about whether the generation model is extracting or copying fragments from the input data, and whether its generations can be considered original. These inquiries target potential intellectual property violations, and copyright infringement, especially if the algorithm's training data is protected by intellectual property policies—which music tends to be. Moreover, originality is often an important criterion in discussions about copyright protection of AI-generated works [123]. While there is significant debate regarding originality, Strum et al. [123] highlights that this characteristic may be established when there is a substantial creative human contribution to the resulting output.

Despite the relevance of originality in copyright legislation and the inquiries into generation algorithms behaviour by different stakeholders, especially artists, assessing the originality of generative algorithms' outputs is narrowly explored in the music domain [134]. We examine some methods to evaluate and assess the originality of generative AI in the next subsection.

Referring to intellectual property rights and regulations, there is scarce literature due to the uncertainty and novel status of AI-generated music, especially in publications focused on music generation algorithms. In our review, we found that intellectual property protections are region-dependent and in many instances, there are no clear decisions in the stipulated legislation. To complement the results on this topic, we briefly explored two of legal publications addressing the copyright law of computer-generated works. For instance, under the UK's *Copyright, Designs and Patents Act (1988)*, section 178 defines "computer-generated" as "generated by computer in circumstances such that there is no human author of the work" [63]. Importantly, section 9(3) of the Act stipulates that the "author" of a computer-generated work is "the person by whom the arrangements necessary for the creation of the work are undertaken." Instead, in other countries without any previous regulation, it is being questioned whether AI-generated works should generate copyright [123]. In the US, the US Copyright Office (USCO) refuses to register the copyright for works "without any creative input or intervention from a human author" [64].

Due to the novelty and rapid growth of music-generative AI, much of the discussion on IP is found mostly in non-scientific platforms, such as blogs and open discussions. Nonetheless, the chapter *Law* in the book *Artificial Intelligence and Music Ecosystem* [39], examines intellectual property and copyright foundation. It points out the challenges associated with copyright law and AI-generated music, acknowledging the role of human creativity. Recently, in a *Communications of the ACM* column, Samuelson [114] introduced generative AI's main legal challenges and questioned the legality of using and deriving outputs from copyrighted data. Further exploration of legal issues derived from generative AI in the music domain is left out of the scope of this review. From 2024, we note a rise in publications addressing the issue of copyright in generative AI and the discussion of authorship: Lee [84], Mantegna [89], Mazzi [90] and Samuelson (2024) [115]. We recommend the indicated publications to better delve into this topic.

Acknowledging the new risks coming along with the advancements in generative AI models, Knott et al. [82] propose that any foundation model must provide a reliable detection mechanism for its generated content, and should be included in current legislation as a requirement for the public release of a model. With a similar perspective, Tang et al. [125] claim that generative AI outputs should be verified from a data management perspective. Intending to resolve this gap, they propose *VerifAI*: a framework designed to verify AI-generated data by assessing its accuracy and consistency against trusted data sources. By comparing generative outputs with known datasets, VerifAI enhances transparency, helping to ensure that generated content meets quality standards and can be reliably used in decision-making processes. Strategies such as *VerifAI* can help to clarify increasing concerns with generative AI trust and reliability.

4.4 RQ$_3$: How can creativity and originality in AI-generated music be evaluated and measured?

Assessing what constitutes creativity remains still a subjective matter with a lack of standardised evaluating strategies [79]. However, there is an increasing trend towards developing objective methods to evaluate human and algorithm creativity objectively [99, 132] For instance, Moruzzi [99] introduces a strategy to evaluate creativity by examining the autonomy, problem-solving, and evaluative capacities of systems, focusing on how generative models such as GANs achieve a level of autonomy that could allow them to be seen as creators. The proposed approach suggests that creativity in music generation can be assessed beyond the final output by considering a model's inherent ability to self-evaluate and navigate creative decisions independently of human intervention. Instead, Agres et al. [6] introduce an overview of current methods to evaluate both human and computer-based creativity, that might be adapted to musical creativity and music metacreation systems. Their work claims the relevance to studying creativity applied to this field, which is still far from being resolved.

Regarding computational creativity evaluation, we found several interesting publications. First, Lamb et al. [83] introduce a detailed survey considering the perspective from multiple disciplines: psychology, philosophy, cognitive science and computer science. With a similar perspective, Mejtoft et al. [91] explore the ability of algorithms to be creative. Their work shows indices of machines being able to achieve human-level complex art and states they should be considered creative entities. Further, it is relevant to note the work by Anna Jordanous, who has made significant contributions to the evaluation of computational creativity [77–79] . The author underscores the need and value to assess creativity in computational systems, identifying key characteristics of creativity, while providing clear definitions to frame creativity evaluation. Jordanous (2011) [78] introduces a set of flexible but specific "Evaluation Guidelines" that encourages objective evaluation among creative systems, providing an understanding of their creative capabilities. Later on, the author introduced the Standardised Procedure for Evaluating Creative Systems (SPECS) [79], which consists of (1) defining what is to be creative for a system, (2) designing tests linked to the previous definition, and (3) conducting those tests. The proposed framework provides insights on how creativity can be measured and compared across systems.

Besides, Natale and Henrickson [100] argue that there is a need to update traditional views on computational creativity to include human perception of the work's creativity. For this reason, they propose the *Lovelace Effect*[17] as an analytical tool to identify when computer-powered works are perceived as creative and original. By shifting focus from the technical capabilities of AI to human interpretations, the *Lovelace Effect* highlights how cultural and social contexts shape perceptions of AI creativity. A notable example is presented in Colombo [42], where they disclose a novel neural network model to automatically compose melodies. They adopt the novelty of musical sequences as a new measure to assess the creativity of their model, which allows them to observe a consistency between the generated melodies and the music style they were trained on.

In the music generation field, Cádiz et al. [47] assess the creativity of two generative models (*TimberNet*[18] and *StyleGAN Pianorrolls*[19]) and demonstrate that these systems were able to learn musical concepts not straightforward in the training data. Thus, they hypothesise the studied generative models may be considered creative. Furthermore, Guo et al. [65] highlight a significant gap in the evaluation of melodies in automatic music generation: while existing methods often rely on subjective human perception, there is a lack of objective methods that can comprehensively assess the quality of generated music. They propose an objective melody evaluation strategy based on music theory and mathematical statistics, which aims to address this gap by providing a more objective assessment of both the generative model and the music it produces. Concurrently, focused on AI-generated music, Xiong et al. [131] present a comprehensive survey on available evaluation strategies for music-generative AI. They differentiate the strategies according to their character: objective, subjective and a combination of both. They review multiple evaluation methods and present a unified reference for music-generative AI evaluation approaches. All of these publications are aligned with the recommendation to further research and address the challenge of evaluating creativity in the music generation field.

For the image generation field, Somepalli et al. [121] focus on analysing whether diffusion models are replicating partial or complete samples from the training data based on object-level similarity. Their findings reveal dataset size and diversity influence replication frequency: more data, less replication. However, this does not imply models trained on large datasets do not copy and/or memorize training data. They demonstrate that the popular model *Stable Diffusion*[20] by Stability AI memorizes and reproduces a statistically significant amount of training data. However, there is scarce literature, centred on music-generative AI, on evaluating what the algorithm is doing, and whether it is replicating the input data or not. This is crucial to determine whether a generated song is original. Despite that, it is not yet clear how

[17]The *Lovelace Effect* is the phenomenon where users perceive computer-generated behaviour as creative and original, regardless of the system's actual autonomy. This concept is conceived as the opposite of the *Lovelace Objection*, which considers computers cannot be creative and only execute given instructions.
[18]https://doi.org/10.1016/j.engstruct.2022.114418
[19]Based on https://ieeexplore.ieee.org/document/9156570
[20]https://stablediffusionweb.com/

generative algorithms should be assessed regarding originality, considering potential risks of plagiarism and copyright infringement [134]. López-Garcia et al. [88], building on their previous work on music similarity measurement of symbolic music, propose *SpectroMap*: an algorithm to estimate music similarity between fragments of digital audio, which could be implemented as an objective tool for automatic plagiarism detection. Alternatively, besides stressing the need for objective evaluation of generative systems, Yang and Lerch [132] present an evaluation method for generative systems of symbolic music based on the assessment of multiple music-informed metrics.

Furthermore, Yin et al. [134] introduce the *originality report*: a methodology to baseline and evaluate how much a generative algorithm reproduces the training data based on two symbolic similarity measurements (cardinality score and fingerprinting score). Their results highlight that the popular model *Music Transformer*[21] copies segments of pieces in the training set.

Some of the novel music generation models, such as *MusicLM* [5] and *MusicLDM* [35], are already acknowledging the issues related to plagiarism, copyright infringement and creative content misappropriation. On the one side, Agostinelli et al. [5] scrutinise training data memorisation by measuring exact coincidences between generated and target tokens on a sampled segment. Their results indicate that only a tiny proportion of generated samples ($< 0.2\%$) are exact matches, which, in general, involve tokens with low diversity. Nonetheless, 1% of the analysed samples exhibit approximate coincidence with the reference samples. Underscoring the need to tackle these risks in future research, they refrain from publicly release the model (*MusicLM*). Alternatively, *MusicLDM* directly addresses these concerns through two beat-synchronous mixup strategies, specifically designed for data augmentation: beat-synchronous audio mixup (BAM) and beat-synchronous latent mixup (BLM). These techniques allow MusicLDM to generate diverse outputs while maintaining musical coherence and minimizing memorization of specific training data. These strategies improve the model's generalisations and reduce overfitting, minimising the number of copied instances in the generations.

In 2024, we identified two publications proposing different approaches to assess the originality of AI-generated music. First, Barnett et al. (2024) [17] introduced a framework to understand training data attribution, based on computing the similarity between training data and AI-generated music. This method computes the cosine similarity between training and generated data samples of two audio embeddings: Contrastive Learning of Musical Representations (CLMR) [122] and Contrastive Language-Audio Pretraining (CLAP) [129]. They validate this methodology applying it to *VampNet* [60], a novel generative model released in 2023, and with a human listening study, to illustrate how similarities between generated and original music can be systematically analysed. This work offers insights into the originality of AI-generated music with a data-driven perspective while assessing training data attribution. Then, Afchar et al. [4] present the first research study focused on detecting music deepfakes successfully, with an accuracy of 99%. However, the authors acknowledge that performance metrics alone are not enough to guarantee the reliability of the model. They underscore that challenges such as robustness to common audio manipulation, generalisation to other generative models, calibration and interpretability must be considered when assessing potentially forged content, like deepfakes.

4.5 RQ$_4$: What is the role of transparency in the assessment of creativity, originality and intellectual property of AI-generated music?

Our last research question proposes to explore investigations related to AI transparency applied in music-generative AI. We aim to review how transparency strategies can contribute to assessing and resolving highly problematic challenges

[21] https://magenta.tensorflow.org/music-transformer

regarding creativity, originality and intellectual property of AI-generated music. From all the set RQs, this is the one with the fewest identified publications. Despite few publications targeting directly specific transparent strategies applied to music generation models, our search led us to interesting approaches and novel research perspectives, which we reflect in this subsection.

AI transparency has a significant role in tackling concerns and challenges towards AI and reducing mistrust in these systems [74]. Many believe its contributions can help generative AI to be more approachable and trustable. For instance, Yin et al. (2022) [134] claim the need to evaluate music-generative algorithms beyond loss and accuracy, to prevent copying training data. A similar point is introduced by Afchar et al. (2024) [4], who emphasise that while music deepfake detection shows promising results, evaluating models should go beyond performance metrics such as accuracy. They argue that addressing broader challenges is critical for future research.

Nonetheless, Barnett (2023) [16] observes that there is minimal research focused on considering the ethical implications of generative audio models. The author presents a systematic literature review of 884 papers, aiming to overview the ethical implications acknowledged by researchers in the field. The findings demonstrate that while 65% of the papers highlight the positive potential of generative audio technologies, less than 10% of the analysed papers discuss the negative ethical implications of these algorithms, such as risks linked to deepfakes and copyright infringement. Barnett underscores the need of researchers to actively consider and address the ethical implications and potential harms associated with generative audio models.

Related to dataset transparency, Morreale et al. (2023) [98] offer a critical analysis of data collection practices in training sets for Automatic Music Generation (AMG) models. Their work systematically reviews datasets used over the past decade in the *International Music Information Retrieval Conference*[22] (ISMIR) publications and reveals that almost half of these datasets were published without explicit permission from musicians, raising significant ethical concerns. This study highlights the lack of proper documentation, particularly regarding how datasets are created, and calls for more responsible and transparent practices in dataset development. However, when we searched and reviewed documentation strategies, we could not find any specific for music-generative models (see Section 4.2.3).

The increasing exploration of co-creation tools is viewed by many as the preferred way to use AI in art generation, arguing there is no art without humans [48]. Indeed, this is an increasing research trend having found many novel research strategies contributing to such initiative. At the same time, different perspectives of human-in-the-loop (HILT) approaches for creative applications are proposed in Chung [36]. They assert: "Advanced HILT approaches are meant to increase interaction between human and AI, and this will help generate more diverse, multimodal, and autonomous systems". Emphasizing that human-AI collaboration is understudied, they argue that computational creativity can be enhanced to be more alike to human creativity. However, in RQ_2 we observed discrepancies in the CC community on the perspective of creativity and the role of algorithms in digital art creation. Thus, the role of AI within music creative practices has multiple approaches and considerations.

Aligned with this perspective, Esling and Davis [53] highlight the complexity of the concept of creativity and discuss the limitations and deficiencies of AI towards this term. The authors intend to align social and computer science perspectives on computational creativity. They emphasise that using computational models for assessing creativity should involve a multidisciplinary approach encompassing social sciences, anthropological studies and algorithmic methods. Daniele and Song [48] also claim to enhance transparency methodologies for computational creativity assessment. In addition, Epstein et al. [52] underscore the significance of transparency, suggesting that a

[22] https://ismir.net/

better understanding of generative AI reinforces AI as a collaborative tool to enhance human creativity, rather than an autonomous agent.

Another observed topic is the relevance of the algorithm errors Baeza-Yates and Estévez-Almenzar [14]. For example, when classifying images of cats and dogs, it is certain for us that a dog assigned the cat category is an error. However, when a music generation algorithm makes a mistake, how can we be certain of that mistake? In other words, what is the relevance of an error in these algorithms? Other investigations in the art domain put this question in perspective. For instance, Caramiaux and Alaoui [31], survey artists who use AI-powered tools and reveal that they often appreciate the errors produced by these systems. Thus, it is complex to determine errors in art generative models only considering loss and accuracy.

It is also interesting to highlight the review presented by Bandi et al. [15], which focuses on the current state-of-the-art in generative AI, covering its requirements, models, input-output formats, and evaluation metrics. Besides providing a comprehensive and detailed study on these aspects, the authors highlight several challenges and implementation issues that generative AI faces, particularly in terms of interpretability, transparency, and training data requirements. This is especially relevant given the ongoing debates around intellectual property and plagiarism in AI-generated content. Additionally, the authors underline the importance of establishing reliable evaluation metrics, which still requires significant research to ensure that generative models can be assessed accurately and fairly across different domains. Furthermore, the review also draws attention to the ethical concerns associated with generative AI, such as its potential misuse for malicious purposes, such as deepfakes. The authors argue that resolving these challenges will require interdisciplinary efforts and call for the establishment of guidelines, regulations, and safeguards to ensure the responsible use of generative models, which is aligned with other publications' perspectives e.g., Cotton and Tartar [46] and Epstein et al. [52]. One of the most significant contributions of this investigation is the presentation of an architecture-based taxonomy of generative AI models, which helps classify models like VAEs, GANs, and diffusion models, among others, aiding researchers and practitioners in selecting the appropriate model for their tasks.

4.6 Limitations

While we aimed our systematic literature review to be comprehensive and inclusive, we acknowledge the particular limitations of our work in this section.

First, since we started our systematic literature review in early 2023, our publication selection date was limited to December 2022. One of the biggest disadvantages of SLR is the constraint of time, as completing all the steps in the process of the review might require an extensive amount of time. However, we recognise the extensive number of articles related to generative AI in the music domain published in 2023 and 2024. To mitigate the time gap between the selection phase and dissemination of the review, we shaped this review to a two-stage review. We coucted a more general transparency in generative AI review until 2022, and then included 34 publications released in 2023 and 2024, carefully selected with a narrow and music-specific perspective.

Another limitation of our review is a bias towards recent publications. Due to the novelty of AI transparency within the music field, which we observe to be scarcely explored, we tend to be particularly inclined towards publications after 2016. This is due to the selected search keywords, which are extremely aligned with AI transparency terminology. Nonetheless, we must acknowledge a long research tradition on computational methods in algorithmic composition and automatic music generation. Moreover, our review focuses predominantly on modern generative AI approaches, particularly those based on ML models, which further narrows our findings towards recent advancements. This focus may overlook contributions from earlier generative algorithms or rule-based approaches that have also influenced music generation and creativity studies. Consequently, our coverage of RQ_4 is constrained to a small number of publications, whose results are not perfectly aligned with the scope of the question. This is consequent to the very little literature available strictly on AI transparency for generative algorithms in the music domain, despite all the indetified claims towards more transparent models.

In addition, we primarily focused on retrieving works from major publication sources (i.e. Scopus, IEEE Xplore, ACM and arXiV), where most computer science and MIR-related publications are typically published. However, they may overlook valuable interdisciplinary perspectives from other academic fields, such as musicology, ethnomusicology, humanities, philosophy and social sciences. At the same time, we limited publications to be available in English, which introduces a bias in our review towards Anglo-centric investigations.

The acknowledged limitations, together with the fact that these topics are now being discussed very actively, suggest an opportunity for future research to update this literature review. Considering this landscape, we have built a dynamic list[23] to continue updating the current collection of 107 publications. By providing a dynamic and public list, we aim to enhance a comprehensive and up-to-date reference space for researchers and practitioners at the crossroad of AI transparency and music generation.

5 DISCUSSION

5.1 Towards transparent music-generative AI

Music-generative AI raises significant challenges, linked to the use of AI tools in music creation processes, the effects on musicians, performers and producers, and intellectual property management. The literature demonstrates that few investigations acknowledge and discuss the negative ethical implications of audio generative models [16]. A significant goal within the AI research community is to understand algorithms' behaviour, analysis and decision-making capabilities to unravel their black-box nature. Consequently, an extensive demand exists to enhance and strive for fair and transparent algorithms [1]. Aligned with this point of view, within the scope of trustworthy AI, transparency is considered a suitable way to evaluate and challenge music-generative AI. Afchar et al. (2024) [4] call researchers to consider and address the challenges derived from novel generative models, beyond assessing them in accuracy and robustness. However, our review reveals a limited number of studies on AI transparency in the music domain, with a few recent initiatives specifically addressing music-generative AI (see Table 4) . Such initiatives yield with the claim to introduce transparency strategies in music generation.

Our review also included publications from other fields, such as image and speech, where AI transparency has become more employed in recent years. Thus, we examine applicable initiatives (e.g., *Datasheet for Dataset*) and suitable approaches in other domains that could be transferred to music generation algorithms (e.g., LIME). Recently released model *MusicGen* includes documentation method *Model Card* in its GitHub repository[24]. Also, Mishra et al. [93] address an explainability approach for music content analysis. Both are clear examples of how a general or specific transparency approach might be applied to the music domain and to music-generative AI. Alternatively, other solutions such as *Focus* [10]—a metric to assess attribution in the image domain—could serve as a reference to enhance the traceability of involved training data in the music generation process.

A significant challenge identified is the inconsistency in the use of AI transparency terminology, with some works treating certain terms as synonyms, such as *explainability* and *interpretability*, while others distinguish between them.

[23] https://roserbatlleroca.github.io/dynamiclist-transparency-music-genAI/
[24] https://github.com/facebookresearch/audiocraft/blob/main/model_cards/MUSICGEN_MODEL_CARD.md

Intending to address this issue, we clarified our interpretations and definitions for the most relevant terms in this domain (Section 2).

Regardless of many voices advocating for the importance of explainable and interpretable methods to achieve more transparency, it is considered a challenging technical issue [2]. Some publications insist that there is a trade-off between a model's accuracy and implementing explainable and interpretable techniques. In that perspective, an interpretable model would not be able to achieve the same performance level as a black-box algorithm. However, multiple examples examined in our review—such as Pintelas et al [106], Böhle and Fritz [24], Zinemanas et al. [137] and Alonso-Jiménez et al. [9]—demonstrate that it is possible to achieve state-of-the-art competitive interpretable models. In addition, Rudin [112] strives to implement directly interpretable models, instead of focusing on developing post-hoc explanations.

Two of the biggest challenges and ongoing discussions regarding AI-generated art are around creativity and originality. The first challenge is determining whether AI-generated works are *creative* and *original*. The second challenge relies whether these works should be considered *creative* and *original*. These issues are directly linked to intellectual property and potential rights violations by algorithms and their outputs. For the music domain, these challenges are extremely relevant because of the role of copyright law within the music industry and the legal and economic consequences of infringing it.

We identify two principal implications for music intellectual property, especially concerning copyright: (i) using copyright material in training data; and, (ii) whether AI-generated music is original and not a direct copy from the training data, and how originality can be evaluated. On the one side, music generation models require large amounts of training data [123], e.g., to produce new songs with similar style, genre or mood. Often copyrighted works are used to train models, which raises concerns about authors' rights violations. Moreover, during the learning process, algorithms may reproduce the training data, which would imply breaking the reproduction right [25]. On the other side, considering the generated outputs, a complex debate appears on whether these systems can be original, or if they are merely replicating from the training data. While generative models are designed to build works that resemble the input data but differ enough to not be replicas of the reference data [47], there is growing concern that AI outputs may still closely resemble specific copyrighted pieces. Replicating from a copyright training set would entail copyright infringement from the model and its output works. Such distinction further complicates when comparing AI to human composers, who also draw inspiration from pre-existing music yet are rarely questioned for originality unless they directly copy. While humans might find inspiration from other artists' works, AI-generated music may be scrutinized more closely due to its reliance on large-scale data and the risk of exact replication without creative intent. This debate—whether AI training and human inspiration should be judged similarly—is provoking discussions and raising questions about how originality should be assessed in AI-generated works.

Nonetheless, as highlighted in RQ_3, scarce literature is focused on evaluating plagiarism in these algorithms and the originality of their outputs [88, 132, 134]. Thus, how to approach and assess originality from the output works - a critical point in copyright regulation - remains uncertain [134]. Regardless of the efforts from regulatory institutions to determine applicable policies, such as the AI Act in the European Union[25], today's regulation is not able to keep up to date with generative models' implications.

Another open question focuses on creativity. We observed how some studies questioned whether algorithms could be creative without human intervention and inquired if the fear of algorithms generating art by themselves is realistic [48]. However, we find two juxtaposed positions, as other works argue that AI has the capability to be creative [12].

[25] https://www.europarl.europa.eu/news/en/headlines/society/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence

This disagreement is acknowledged too by Moruzzi [99]. A complementary perspective is offered by Epstein et al. [52], who argue that AI generative tools do not replace human creativity but rather provide a distinct medium that requires human intent and oversight to achieve meaningful creative outcomes. They suggest that AI tools should function as facilitators, as another creative instruments, emphasizing *"meaningful human control"* as essential for genuine creativity.

Multiple questions appear around the role of AI in digital art creation and the development of creative tasks. Currently, much of the literature focuses on the potential of co-creativity with AI tools, emphasizing collaboration between humans and AI rather than AI being creative without human intervention. This trend can be observed in the rise of music-generative tools that support and intend to enhance human creativity. However, an alternative research direction—highly supported by the industry—explores AI systems that require less human involvement, such as text-to-music initiatives, where an algorithm can generate music tracks directly from an input text description. Nonetheless, how to properly evaluate such a subjective topic as creativity remains an unresolved inquiry. Even if research on objective evaluations has increased, it is complicated to assess art generation without subjective perspectives.

Despite some of the identified publications acknowledging the need for transparency to understand music-generative algorithms, evaluating the creativity and originality of these models and their generated outputs remains an open issue. Our review underscores a lack of AI transparency strategies focused on music-generative models, addressing their ethical, social, legal and economic implications.

5.2 Main identified research gaps and challenges

Our main goal in conducting a systematic literature review on transparency in music-generative AI was to identify relevant research gaps and challenges in this complex and novel field. After discussing the results retrieved from our research questions, we subsequently present a list of notable research gaps and challenges detected. In addition, we present a visual summary of them in Figure 5.

The main identified challenges and research gaps of transparency in music-generative AI are:

- **Lack of standard terminology for AI transparency**: There is no consensus on AI-transparency terminology, leading to key terms being used inconsistently across publications. We outlined our perspective on AI transparency terminology in Section 2. Nonetheless, the community is still far from achieving a standardization of vocabulary definitions.
- **Technical complexity of AI transparency**: While transparency techniques are rising, many publications highlight significant technical challenges to be addressed. We observe the increasing tendency of techniques to resolve the "black-box mystery", but further development is needed. Some identified publications demonstrate that interpretable by-design models can be competitive as black-box models in terms of accuracy and task performance. Thus, there is a wide range to enhance research in this area.
- **Lack of music-specific approaches in AI transparency**: The limited number of transparency-focused identified publications for music-generative AI demonstrate a gap in the interaction between AI transparency and music-generative AI. Both fields have expanded and gained much popularity in recent years, but their alliance is still scarcely explored. Thus, there is a need to adapt and design novel methods for the music domain.
- **Lack of standard methodology to objectively evaluate AI-creativity**: Even if creativity is a widely studied topic, there are different perspectives and interpretations on what is creativity, what is being creative, and what is understood and constitutes both human and machine creativity. This subjectivity presents a challenge in developing standardised, objective methodologies for evaluating AI-driven creativity. The main difficulty lies in

establishing criteria that can assess creativity in AI-generated music objectively and without relying only on human judgment, which varies across artistic and cultural contexts.

- **Ambiguity in the role of AI in creative processes**: There is an ongoing debate about the role of AI in creative processes. Some studies emphasise AI's role as a supportive tool that enhances human creativity, often through co-creation tools that involve collaboration between artists and AI. In contrast, other perspectives suggest that AI can independently generate artwork and, therefore, possesses the capability of being creative. This ambiguity in the role of AI raises questions about potential shifts and conditioning in creative processes traditionally undertaken only by humans.
- **Lack of methods for assessing originality in music-generative AI**: Despite the advancements in generative modes, there remains a significant gap in the literature regarding effective methods for evaluating whether AI-generated music replicates elements from its training data. This issue is crucial because it affects the trustworthiness of generative AI and the originality of its outputs. Without reliable methods to assess this, there are serious implications for intellectual property rights and potential copyright violations. To address this, there is a pressing need for more research into techniques that can accurately determine whether AI-generated music is original or reproducing elements from the training data.
- **Substantial ambiguity in the intellectual property of AI-generated works**: Regulation differences across countries and limited understanding of system behaviours create uncertainty about potential intellectual property violations and the criteria of copyright law on AI-generated works. This is of great significance in the music domain, considering all the legal and economic implications of today's music industry.

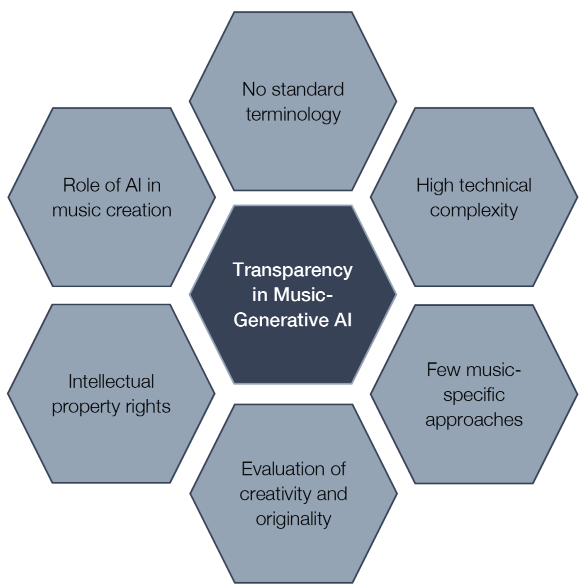

Fig. 5. Main identified challenges around transparency in music-generative AI.

6 CONCLUSION

In this work, we aimed to examine the literature on AI transparency in the context of music-generative AI, focusing on the connections between transparency, artificial intelligence, music generation, the evaluation of creativity and originality and the connections to intellectual property rights. We presented a comprehensive two-stage review of 107 identified publications, highlighting an increasing interest in AI transparency and the ethical consequences of generative models. Nonetheless, transparency strategies applied to music-generative AI are significantly under-explored. Furthermore, our review points out several disagreements within the identified publications, especially concerning creativity, music-generative AI and the role of algorithms in music-creative tasks.

One of the key contributions of this review is the identification of research gaps and challenges in the interaction of transparency and generative AI in the music domain (Section 5.2). Overall, the present review provides relevant information for researchers, particularly in the MIR field, and practitioners in AI transparency and music generation. We recommend Section 2 to grasp the key vocabulary and definitions in AI transparency. Subsequently, RQ_1 can be especially useful for readers getting started in AI transparency, as it reviews various strategies. Alternatively, RQ_4 highlights the lack of investigations applying transparency approaches in music-generative AI.

Our findings suggest that implementing transparency methodologies within music-generative AI can help address multiple concerns and challenges identified in this study. For instance, such strategies can contribute to determining if a piece of AI-generated music is novel and original, assessing whether intellectual property rights associated with the training data have been violated, and clarifying the role and involvement of AI in the music creation process. Thus, despite the scarce exploration of these approaches, there is significant potential for blending transparency strategies with music-generative AI. We observe research trends that reveal this topic is rapidly gaining popularity and attention. To further support ongoing developments, we have created a dynamic list in a public repository, aiming to encourage updates to this review and stimulate continued research in this area. With this review, we hope to encourage and enhance the development of transparency methodologies in music-generative AI.

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